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ORIGINAL ARTICLE

On Probability Estimation for Unbalanced Classification

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Abstract

Unbalanced classification poses a significant challenge in real-world applications such as medical diagnosis and fraud detection, where class label distributions are highly skewed. Standard classification methods often yield suboptimal results by sacrificing minority class accuracy to maximize overall performance. Common techniques to address unbalanced classification typically involve weighting adjustments of points for different classes or resampling strategies, including oversampling of minority classes and undersampling of majority classes. While these balancing techniques are effective in improving classification accuracy, they often introduce biases that compromise posterior class probability estimation and model calibration. This paper highlights the trade-offs associated with balancing techniques when applied without appropriate adjustments. We systematically investigate the distortions in probability estimation caused by these unbalanced classification techniques and propose a robust framework to correct these biases through probability adjustment. We further investigate high-dimensional unbalanced data, examining both the distortion induced by balancing techniques and the shrinkage effect of regularization on probability estimation. Finally, our method is evaluated through simulations and real data analysis, covering a range of balancing strategies including class weighting, random resampling, and generative models. Results demonstrate that our proposed framework significantly mitigates probability estimation bias while preserving classification power when applying traditional balancing techniques, ensuring reliable posterior estimates.

Keywords Calibration · Machine Learning · Oversampling · Undersampling · Weighting

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1 Introduction

Class imbalance is prevalent in many real-world classification problems, where one class significantly outnumbers the others. This phenomenon is often unavoidable in many real-world scenarios. For example, rare diseases occur much less frequently than common health conditions in medical diagnostics [1]. Similarly, fraudulent transactions are far rarer than legitimate ones in fraud detection [2, 3].

Class imbalance poses a significant challenge for standard classification algorithms in supervised learning, which typically assume an approximately equal proportion of classes. When class proportions are highly skewed, standard methods tend to favor the majority class, resulting in poor performance for the minority class. For example, consider a dataset with 1000 instances, where 950 belong to the positive class and 50 to the negative class. A model that naively predicts all instances to the positive class would achieve an overall high classification accuracy of $\frac{950}{1000} = 95\%$. However, such a model would fail to identify any instances of the minority class, making it ineffective for detecting minority-class instances.

To address this issue, various techniques have been developed in the literature to balance class distributions during training. Common strategies include oversampling of the minority class, undersampling of the majority class, and class weighting. Oversampling increases the representation of the minority class by duplicating existing samples or generating synthetic ones [4–6]. More recently, deep generative models have been used to synthesize minority-class observations, providing a model-based approach to synthetic-data augmentation [7, 8]. Undersampling reduces the dominance of the majority class by removing some of its instances [9, 10]. Class weighting modifies the loss function to assign a higher loss to misclassifications of the minority class [11–13]. These methods have demonstrated improvements in classification performance on unbalanced datasets, as reflected in metrics such as Area Under the Receiver Operating Characteristic Curve (AUC), Geometric Mean (G-Mean), and the F1 score [7, 14–17].

While balancing techniques often improve classification performance by mitigating majority-class dominance, they implicitly alter the class distribution in the training data. As a consequence, the posterior probability estimates $P(y \mid x)$ obtained from balanced data may become distorted. To illustrate potential issues with balancing techniques, we construct a toy dataset with five features in which only 1% of the samples belong to the positive class. We train two logistic regression models: one on the original unbalanced data and the other on a version rebalanced via random oversampling. As illustrated in Figure 1, oversampling shifts the estimated probabilities from the extremes (0 and 1) toward the decision boundary (0.5), artificially inflating the perceived uncertainty. Consequently, the fitted model fails to reflect the ground truth, in which the positive and negative classes are in fact well separated.

Distortions in probability estimation can lead to suboptimal decisions in critical applications, such as estimating disease probability in clinical diagnostics or evaluating credit risk in finance. A natural approach is to use two separate models: one trained on the original unbalanced data for probability estimation and another trained on the artificially balanced data for classification. However, this strategy can lead to inconsistent predictions. Since balancing techniques modify the decision boundaries, the new

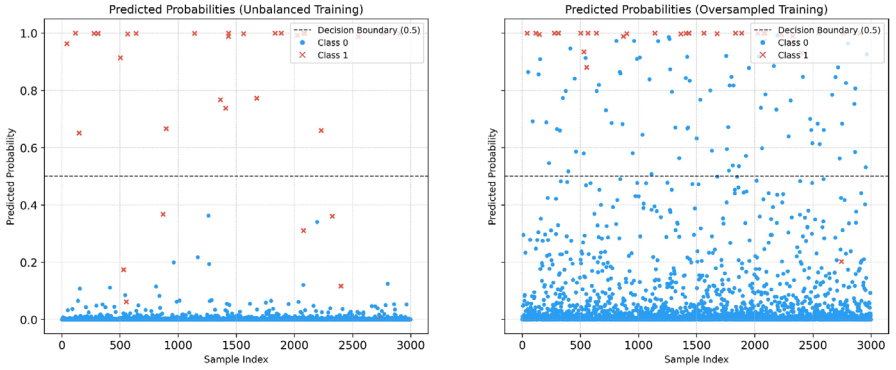


Fig. 1 Distributions of estimated posterior probabilities from models trained on unbalanced data without and with the random oversampling technique. Compared to the unadjusted model, the distribution of estimated posterior probability from the oversampled version has a higher proportion of cases around 0.5

classification model may produce labels that are inconsistent with those generated by the original model. Such discrepancies can be especially problematic in applications that rely on coherent probability-classification alignment for decision-making. Therefore, it is critical to evaluate how balancing techniques influence probability estimation and to develop methods that retain the benefits of balancing techniques to classification while addressing the probability estimation bias caused by these techniques.

Several studies have observed this phenomenon and proposed partial remedies [18, 19]. However, a unified understanding of how different balancing strategies affect probability estimation, together with a general correction framework, remains lacking. In this paper, we systematically investigate the probability estimation biases introduced by various balancing methods and propose a unified framework that improves probability calibration while preserving classification performance.

In this paper, we propose methods for obtaining accurate posterior probability estimates for a broad class of balancing strategies, including oversampling, undersampling, weighting, and synthetic data augmentation. We first identify how these techniques distort posterior probabilities and introduce corresponding adjustments to restore accurate probability estimates. Our approach can retain the classification benefits of balancing methods in unbalanced learning while ensuring estimated posterior probabilities remain suitable for decision-making. The analysis further extends to high-dimensional settings, where probability estimation is distorted by both balancing techniques and the shrinkage bias of regularization. Through comprehensive simulations and real data experiments under both traditional and generative balancing schemes, we show that our proposed framework effectively reduces probability estimation bias, maintains high classification accuracy, and improves the reliability of probability estimates in real-world applications.

Before presenting the details of our proposed framework and its associated results, we first review existing balancing techniques for binary classification in Section 2, followed by a discussion of related work in Section 3.

2 Unbalanced Classification

Unbalanced classification has long been a challenge in machine learning due to its tendency to produce classifiers that favor the majority class, often resulting in poor performance on the minority class [20–23]. Traditional classification algorithms generally assume a relatively balanced class distribution and aim to minimize overall misclassification error, rather than addressing errors within each class. As a result, models trained on unbalanced data may achieve high overall accuracy while significantly underperforming for the minority classes.

2.1 Oversampling

Oversampling is one of the most common strategies used to address the challenges posed by unbalanced datasets. The core idea of oversampling is to increase the size of the minority class, either by replicating existing samples or generating new synthetic ones, thereby achieving a more balanced class distribution. This can be implemented through techniques such as random oversampling and Synthetic Minority Oversampling Technique (SMOTE)-based methods [4].

Random oversampling involves duplicating randomly selected samples from the minority class until the desired class balance is achieved. While straightforward and easy to implement, random oversampling can lead to overfitting, as it simply replicates original samples. This causes the model to become overly familiar with repetitive data points, which can degrade its generalization capability [24].

SMOTE is an improved oversampling approach introduced by Chawla et al. [4]. SMOTE generates synthetic samples by identifying the k-nearest neighbors for each minority class instance and creating new samples along the line segments between the instance and its neighbors. This process increases the representation of the minority class while reducing the risk of overfitting, as the generated samples are not mere duplicates of the original data but interpolations that introduce new diversity. Subsequent variations of SMOTE have been proposed to further refine the oversampling process. Borderline-SMOTE [5] focuses on generating synthetic samples for minority class instances that lie near the decision boundary, which are more likely to be misclassified. Adaptive synthetic sampling [6] adapts the number of synthetic samples generated for each instance based on its local density, prioritizing regions with higher classification difficulty. These variations aim to enhance the effectiveness of oversampling by targeting challenging areas within the feature space, improving both classification accuracy and generalization [25].

2.2 Undersampling

Undersampling addresses unbalanced data by reducing the majority class size. Common undersampling methods include random undersampling and cluster-based approaches [9, 10, 26].

Random undersampling involves randomly selecting and discarding instances from the majority class. While straightforward, the randomness of the selection process can

result in the removal of informative instances, which may degrade the model's ability to generalize from the training data to unseen data in real-world scenarios.

Cluster-based undersampling (CUS) is an alternative developed to mitigate the limitations of random undersampling. These methods cluster the majority class and then either select representative instances or reduce the sizes of each cluster. By preserving the internal aggregate-level structure of the majority class, cluster-based undersampling aims to maintain the dataset's informational integrity and support better generalization in classification tasks. For example, the CUS algorithm proposed by Cao et al. [27] clusters the majority class, identifies informative support vectors using the Support Vector Machine (SVM), and eliminates non-support vectors to reduce the unbalance ratio. Tang et al. [28] introduced granular support vector machines to iteratively retain informative local support vectors within each granule of the majority class, and Tang et al. [29] later extended the method to further refine local support vector selection. These approaches improve the model's focus on the most relevant majority class instances, enhancing classification performance for minority classes. However, this still comes at the cost of losing some information due to discarding data [30].

2.3 Weighting

In addition to sampling techniques such as oversampling and undersampling, weighting is another popular solution to address unbalanced classification by modifying the loss function during training. Specifically, the loss contributions from minority class instances are amplified by assigning them higher weights, while those from majority class instances may be reduced by assigning them lower weights. This adjustment encourages the learning algorithm to prioritize accuracy for the minority class by penalizing errors more heavily for that class [11, 12]. For example, Qiao et al. [31] studied the weighting strategy for margin-based classifiers to deal with unbalanced multicategory classification. Steininger et al. [13] addressed this issue by proposing DenseWeight, a sample-weighting approach based on kernel density estimation, and DenseLoss, a cost-sensitive learning method. These methods dynamically adjust the contribution of each instance to the loss function, ensuring better alignment with the model's learning objectives. They provide a flexible mechanism to prioritize minority class performance without altering the class distribution directly.

2.4 Synthetic Data Augmentation

With the emergence of complex, high-dimensional datasets, traditional oversampling methods can easily lead to overfitting or degraded performance by relying too heavily on limited intra-class information. Although synthetic data generated by traditional methods such as SMOTE can partially remedy this issue, these methods often fail to capture the complex pattern of modern datasets. In contrast, deep generative models have been shown to be powerful in learning distributions of high-dimensional data; consequently, synthetic data augmentation from such models has been a competitive choice for oversampling [7].

Most existing approaches train a generative model to learn the conditioning distribution of observations given its label, such as the conditional Generative Adversarial Network (cGAN, [32]) or variational autoencoder (cVAE, [33]), and then oversample the minority class using synthetic data from these models [8, 34]. These approaches are widely applied across various domains, including image, text, and tabular classification tasks. Moreover, deep generative augmentation can be integrated with classical framework such as SMOTE to form hybrid oversampling schemes that further improve minority-class representation [35].

3 Related Work

Most existing studies on unbalanced classification have adopted balancing techniques, such as oversampling, undersampling, and class weighting, as practical solutions to reduce classifier bias toward the majority class [36–38]. These methods have proven effective in shifting model focus toward the minority class, often improving classification metrics such as AUC, F1-score, and G-Mean [14–17]. Due to this practical effectiveness, many empirical studies adopt such methods without further adjustment.

While these balancing techniques can improve classification outcomes, they often overlook the distortion they introduce to posterior class probability estimation. Rebalancing methods, including oversampling, undersampling, and synthetic data augmentation, alter the prior class distribution presented to the learning algorithms. Classifiers trained on the rebalanced data perceive the modified class proportions as true population priors. Consequently, the class conditional probabilities estimated by these models no longer accurately represent the true underlying distributions, but rather reflect the artificially balanced scenarios [19]. Similarly, weighting modifies the loss function such that the optimization process treats reweighted sample contributions as if they reflect the actual class proportions. This implicitly redefines the effective class priors, leading to biased log-odds estimates [11, 31], and, consequently, distorts posterior probability estimates [11, 18]. In particular, these effects cause estimated probabilities to be compressed toward the center, reducing the separability between the classes in cases where the true posterior should be more polarized (e.g., closer to 0 or 1) in unbalanced data.

Distortions in estimated probabilities can be problematic for real-world scenarios where accurate probability estimation is crucial [39], such as medical diagnostics, fraud detection, and risk prediction [2, 3, 16]. Some efforts have been made to address the distortion in probability estimation introduced by balancing techniques. Dal et al. [19] calibrated posterior probabilities under undersampling using empirical correction. Zhang et al. [18] studied how regularization causes distortion in high-dimensional settings. However, these approaches typically lack general correction formulas and do not provide a unified adjustment framework across different rebalancing methods. Moreover, post-hoc calibration methods such as Platt scaling and isotonic regression [40] do not correct the structural distortion caused by sampling or weighting. Therefore, an additional correction step is required to realign the estimated probabilities with the true, unbalanced population distribution.

Our work systematically derives closed-form corrections for the distorted posterior probabilities resulting from oversampling, undersampling, weighting, and synthetic data augmentation. We further unify these corrections under a single framework and link them with optimal threshold adjustment. Table 1 summarizes the key characteristics of existing solutions and our proposed method, with a focus on their impact on posterior probability estimation. Section 4 introduces a principled adjustment framework that fulfills this objective while retaining the classification benefits brought by balancing strategies. Despite its simplicity, our proposed adjustment is shown to be effective by our numerical studies in Sections 5 and 6.

4 Theoretical Framework

This section elucidates the mechanisms by how balancing strategies distort posterior probability estimation and identifies the resulting shifts in optimal decision boundaries. We derive closed-form correction formulas for resampling and weighting, and extend this analysis to high-dimensional settings where regularization-induced shrinkage compounds these biases.

4.1 Posterior Shift After Balancing

Consider a binary classification problem with classes $y = 1$ (minority class) and $y = -1$ (majority class), where $\pi_+ = P(y = 1)$ and $\pi_- = P(y = -1)$ represent the prior probabilities of the minority and majority classes, respectively. Using the Bayes' theorem, the posterior probability before applying any balancing technique is:

$$\eta(x) = P(y = 1 \mid x) = \frac{\pi_+ P(x \mid y = 1)}{\pi_+ P(x \mid y = 1) + \pi_- P(x \mid y = -1)}, \tag{1}$$

where $P(x \mid y = 1)$ and $P(x \mid y = -1)$ are the likelihoods of observing the data point x given that it belongs to the minority and majority classes, respectively.

4.1.1 Posterior Shift from Resampling

Our analysis assumes that resampling procedures modify the empirical class priors while leaving the class-conditional distributions approximately unchanged. After applying resampling or synthetic data augmentation, the class priors change from π_+, π_- to π'_+, π'_-, while the likelihood $P(x \mid y)$ remains the same. Under this assumption, the posterior probability after balancing becomes

$$\eta'(x) = \frac{\pi'_+ P(x \mid y = 1)}{\pi'_+ P(x \mid y = 1) + \pi'_- P(x \mid y = -1)}. \tag{2}$$

For simplification, let $a(x) = P(x \mid y = 1)$, $b(x) = P(x \mid y = -1)$, and $c(x) = \frac{b(x)}{a(x)}$. Since $c(x)$ represents the intrinsic ratio of likelihoods, it remains constant regardless

Table 1 Comparative analysis of common solutions to unbalanced classification and the proposed correction method

Method	Advantages	Drawbacks	Effect on Posterior Probability
Unbalanced training	- Reflects true class priors - No information loss	- Poor minority performance - Biased decision boundary	- Well-calibrated estimates
Oversampling and synthetic	- Improves minority recall - Preserves all data	- Overfitting risk - Increased training time	- Inflates the minority prior distribution - Overestimates minority posterior probability
Undersampling	- Simple to implement - Reduces majority bias	- Information loss - Unstable decision boundary	- Deflates the majority prior distribution - Underestimate majority posterior probability
Weighting	- Uses full data - Tunable loss emphasis	- Sensitive to scale	- Alters log-odds - Miscalibrated posterior probabilities
Proposed method	- Retains classification gains - Corrects probability bias	- Additional post-processing	- Realigns with true posterior probability - Improves calibration

of balancing. Substituting $c(x)$ into (1) and (2), we have $\eta(x) = \frac{\pi_+}{\pi_+ + c(x)\pi_-}$ and $\eta'(x) = \frac{\pi'_+}{\pi'_+ + c(x)\pi'_-}$. By expressing $c(x)$ in terms of $\eta(x)$, we can get $c(x) = \frac{\pi_+ - \eta(x)\pi_+}{\eta(x)\pi_-}$. Then the posterior probability after resampling is:

$$\eta'(x) = \frac{\pi'_+}{\pi'_+ + \frac{\pi_+ - \eta(x)\pi_+}{\eta(x)\pi_-}\pi'_-} = \frac{\eta(x)\pi'_+ - \eta(x)\pi_+\pi'_+}{\pi_+ - \eta(x)\pi_+ + \eta(x)\pi'_+ - \pi_+\pi'_+}. \tag{3}$$

This demonstrates how resampling techniques influence posterior probabilities by modifying class priors.

For oversampling and synthetic data augmentation, let $\alpha \geq 1$ denote the oversampling ratio, which specifies the factor by which the number of minority class samples is increased. Typically, for a perfectly balanced dataset, we set the balanced prior probabilities to be equal with $\pi'_+ = \pi'_- = 0.5$. Utilizing (3), we express $\eta(x)$ in terms of $\eta'(x)$:

$$\eta(x) = \frac{\eta'(x)}{\alpha + \eta'(x)(1-\alpha)}. \tag{4}$$

Similar to oversampling, we let β represent the undersampling ratio, which indicates the probability of retaining a majority class sample during undersampling. With $\beta \leq 1$ and $\pi'_+ = \pi'_- = 0.5$, we then express $\eta(x)$ in terms of $\eta'(x)$ based on (3):

$$\eta(x) = \frac{\beta\eta'(x)}{1 + \eta'(x)(\beta - 1)}. \tag{5}$$

4.1.2 Posterior Shift from Weighting

For the unweighted logistic regression, the model minimizes the logistic loss: $\ell(y, f(x)) = \log(1 + \exp(-yf(x)))$ [41]. The minimizer satisfies: $f^*(x) = \log\frac{\eta(x)}{1-\eta(x)}$. With class-specific weights w_+ for class $y = +1$ and w_- for class $y = -1$, the weighted loss is: $\ell_w(y, f(x)) = w_y \log(1 + \exp(-yf(x)))$. Then the expected weighted population risk becomes:

$$R_w(f(x)) = w_+\eta(x)\log(1+\exp(-f(x))) + w_-(1-\eta(x))\log(1+\exp(f(x))). \tag{6}$$

Taking the derivative with respect to $f(x)$ and setting the derivative equal to zero, we have the new minimizer for weighted logistic regression: $f^*_w(x) = \log\frac{w_+\eta(x)}{w_-(1-\eta(x))}$. Applying the sigmoid function $\sigma(z) = \frac{1}{1+\exp(-z)}$, which maps log-odds to probabilities in the interval [0, 1], to this minimizer yields the posterior probability after weighting: $\eta'(x) = \sigma(f^*_w(x)) = \frac{w_+\eta(x)}{w_+\eta(x) + w_-(1-\eta(x))}$. In order to correct the distortion, we express $\eta(x)$ in terms of $\eta'(x)$:

$$\eta(x) = \frac{w_-\eta'(x)}{w_-\eta'(x) + w_+(1-\eta'(x))}. \tag{7}$$

These correction formulas provide a theoretical basis for adjusting posterior probabilities after applying balancing techniques, ensuring accurate probability estimation while benefiting from improved classification performance.

4.1.3 Unified Correction Formula

The corrections above for oversampling, undersampling, weighting, and synthetic data augmentation can be unified using a general adjustment parameter $\kappa > 0$, defined as the relative scaling of the majority class with respect to the minority class. Specifically, κ corresponds to $\frac{1}{\alpha}$ for oversampling and synthetic data augmentation, β for undersampling, and $\frac{w_-}{w_+}$ for weighting. The parameter κ encapsulates the relative prior belief adjustment. When $\kappa < 1$, the correction counteracts the artificial inflation of the minority class, effectively pushing the boundary back to reflect the true rare-event nature of the population. Thus, the corrected posterior probability is:

$$\eta(x) = \frac{\kappa \cdot \eta'(x)}{1 + \eta'(x) \cdot (\kappa - 1)}. \tag{8}$$

This unified formula provides a coherent adjustment framework across different balancing strategies, simplifying both implementation and theoretical analysis.

4.2 Decision Threshold Choice

After correcting the posterior probabilities, it is essential to choose the decision threshold τ, which determines how predicted probabilities are translated into class labels. The default threshold is typically $\tau = 0.5$, representing no bias toward either class. However, this default threshold may not be optimal for unbalanced datasets. Adjusting the threshold can account for differences in class distributions, reduce misclassification costs associated with the minority class, and improve overall model performance.

In the standard case where misclassification costs are equal, the loss function is defined using 0-1 loss. The objective is to minimize the expected risk R, expressed as $R = \mathbb{E}_{x,y}[\ell(y, f(x))]$, where $\ell(y, f(x))$ represents the loss when the predicted label is $\hat{y} = \text{sign}(f(x))$ and the true label is y. The decision function $f(x)$ maps input x into $\mathbb{R}$, and $\text{sign}(f(x)) = 1$ if $f(x) \geq 0$, and $\text{sign}(f(x)) = -1$ if $f(x) < 0$. Then the expected risk can be expanded as $R = \mathbb{E}_x \left[\mathbb{E}_{y|x}[\ell(y, f(x)) \mid x]\right] = \mathbb{E}_x \left[(1 - \eta(x)) \cdot \ell(-1, f(x)) + \eta(x) \cdot \ell(1, f(x))\right]$.

Since the true distribution $P(x)$ is typically unknown, computing R directly is infeasible. Instead, empirical risk minimization approximates R using the empirical distribution of the data. From Table 2, only incorrect classifications are penalized in typical classification problems, leading to the empirical risk formula:

$$\hat{R} = \frac{1}{n} \sum_{i=1}^{n} \left[\ell_{\text{FP}} \cdot \mathbb{I}(y_i = -1, \hat{y}_i = 1) + \ell_{\text{FN}} \cdot \mathbb{I}(y_i = 1, \hat{y}_i = -1)\right], \tag{9}$$

Table 2 Loss Matrix for Binary Classification with 0–1 Loss

	Predicted Positive (+)	**Predicted Negative** (−)
Actual Positive (+)	True Positive ($\ell_{TP} = 0$)	False Negative ($\ell_{FN} = 1$)
Actual Negative (−)	False Positive ($\ell_{FP} = 1$)	True Negative ($\ell_{TN} = 0$)

where $\mathbb{I}(\cdot)$ is the indicator function. Thus, for any instance i, the risks associated with predicting the positive or negative class are $R^+(x_i) = \ell_{FP} \cdot (1 - \eta(x))$ and $R^-(x_i) = \ell_{FN} \cdot \eta(x)$.

The model predicts the class with the lower risk. If $R^+(x_i) \leq R^-(x_i)$, the model predicts the positive class. This leads to $\eta(x) \geq \frac{\ell_{FP}}{\ell_{FP}+\ell_{FN}}$, which defines the decision threshold τ:

$$\tau = \frac{\ell_{FP}}{\ell_{FP} + \ell_{FN}}. \tag{10}$$

In the standard case where misclassification costs are equal ($\ell_{FP} = \ell_{FN} = 1$), the threshold simplifies to $\tau = 0.5$, reflecting no preference between classes.

4.2.1 Threshold After Class Weighting

In unbalanced datasets, a weighted loss function is often employed to assign different importance to classes: $\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} w \cdot \ell(y_i, f(x_i))$. Using (9), the weighted empirical risk is:

$$\hat{R}_w = \frac{1}{n} \sum_{i=1}^{n} \left[w_- \ell_{FP} \cdot \mathbb{I}(y_i = -1, \hat{y}_i = 1) + w_+ \ell_{FN} \cdot \mathbb{I}(y_i = 1, \hat{y}_i = -1) \right],$$

where w_- and w_+ are weights for the majority and minority classes. Typically, these weights are inversely proportional to class proportions with $w_+ = \frac{1}{\pi_+}$ and $w_- = \frac{1}{\pi_-}$.

Similar to the standard case, where the model will predict the class label with the lower risks, the decision rule becomes $w_- \ell_{FP} \cdot (1 - \eta(x)) \leq w_+ \ell_{FN} \cdot \eta(x)$. In equal cost case, this yields the weighted threshold τ_w:

$$\tau_w = \frac{w_-}{w_- + w_+} = \pi_+. \tag{11}$$

When the minority class is underrepresented, π_+ is small, lowering the threshold enhances the identification of minority class instances and consequently reduces false negatives.

4.2.2 Threshold After Resampling

Oversampling, undersampling, and synthetic data augmentation adjust class frequencies to balance the dataset, thereby implicitly assigning different weights to the classes. While resampling techniques modify the empirical distribution of the training data and

weighting directly modifies the loss function, both approaches influence the decision boundary in a similar manner. Consequently, despite their different implementations, these methods all aim to balance the effective contribution of each class during training, and oversampling or undersampling can be viewed as approximate special cases of class weighting.

In the context of oversampling and synthetic data augmentation, we can interpret the oversampling ratio α as an approximation to the weight assigned to the minority class denoted as w_+. Similarly, the undersampling ratio β can be considered as w_-. Based on these relationships, the decision thresholds τ_o for oversampling and τ_u for undersampling can be expressed as follows:

$$\tau_o = \frac{1}{1+\alpha} = \frac{1}{1+\frac{\pi_-}{\pi_+}} = \pi_+, \tag{12}$$

$$\tau_u = \frac{\beta}{1+\beta} = \frac{\frac{\pi_+}{\pi_-}}{1+\frac{\pi_+}{\pi_-}} = \pi_+. \tag{13}$$

These formulations confirm that the thresholds for oversampling and undersampling align with the proportion of the minority class, ensuring consistent treatment across balancing strategies.

4.3 Classification Accuracy After Correction

To correct the distorted probability estimates induced by balancing, we define a unified strictly increasing calibration function $g_\kappa(x)$ based on the adjustment parameter κ, as expressed in Equation (8), where $\kappa = \frac{1}{\alpha}$ for oversampling, $\kappa = \beta$ for undersampling, and $\kappa = \frac{w_-}{w_+}$ for weighting. We define the corrected probability and threshold as:

$$\eta^*(x) = g_\kappa(\eta'(x)), \quad \tau^* = g_\kappa(\tau) = g_\kappa(0.5) = \pi_+.$$

The final threshold satisfies $\tau^* = \pi_+$, aligning with the optimal thresholds derived in (11)–(13). Because g_κ is strictly increasing, it preserves decision boundaries:

$$\eta^*(x) \geq \pi_+ \iff \eta'(x) \geq 0.5.$$

Thus, the corrected classification rule $\widehat{y}(x) = \mathbb{I}\{\eta^*(x) \geq \pi_+\}$ produces exactly the same labels as the rule based on $\mathbb{I}\{\eta'(x) \geq 0.5\}$. This property ensures that the proposed probability recalibration is label-invariant, allowing practitioners to recover reliable uncertainty estimates without compromising the predictive power gained from balancing techniques.

5 Simulation Studies

In this section, we first illustrate the effect of adjustment on probability estimation using toy examples in Section 5.1. Then we present three different simulation scenarios to further compare different methods in Section 5.2.

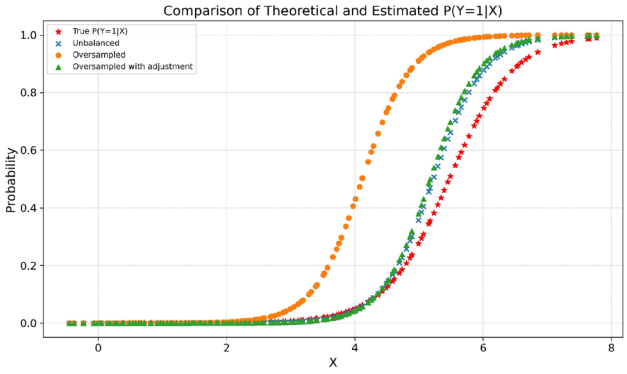


Fig. 2 Comparison among estimated posterior probabilities from models trained on original unbalanced data, oversampled data, and oversampled data with correction, against the theoretical true posterior probabilities

5.1 Illustration of Posterior Probability Adjustment

To illustrate the difference between estimated probabilities and the true posterior probability, we construct a simple synthetic dataset consisting of a single covariate X. We generate a training set of size 1,000 and a test set of size 5,000 from the same data-generating process. The class label $Y \in \{0, 1\}$ follows a Bernoulli distribution with $P(Y = 1) = 0.05$. Conditional on Y, the feature X follows a normal distribution: $X \mid Y = 1 \sim \mathcal{N}(5, 1)$ and $X \mid Y = 0 \sim \mathcal{N}(3, 1)$. We obtain the estimated posterior probabilities by training a logistic regression model on (i) the original unbalanced dataset, (ii) the oversampled balanced dataset, and (iii) the oversampled dataset with our proposed probability correction. For comparison, we also compute the ground-truth posterior probabilities. As shown in Figure 2, oversampling introduces visible distortion—compressing estimation toward 0.5, while our proposed correction adjusts the distorted estimates closer to the true posterior. The corrected probabilities realign with those from the original unbalanced model, effectively mitigating the probability bias introduced by oversampling.

To illustrate how our correction adjusts the posterior probabilities after oversampling, we present the distribution of estimated $P(Y = 1 \mid X)$ before and after correction in Figure 3. The recalibrated probabilities become more concentrated near 0, reflecting the inherent structure of the original unbalanced data, where most instances belong to the majority class ($Y = 0$). This shift corrects the artificial flattening effect introduced by oversampling and restores the probability distribution to better match the true underlying distribution. The resulting estimates are thus more representative of the true class distribution and more suitable for downstream tasks that rely on well-calibrated probabilities.

5.2 Numerical Experiments

Next, we present three simulation scenarios of increasing complexity to assess the effectiveness of our balancing and posterior-probability correction methods. For all

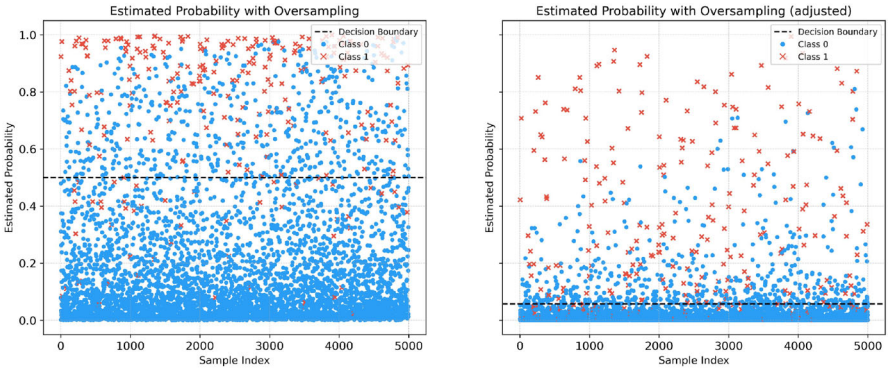


Fig. 3 Distributions of estimated posterior probabilities before and after correction for an oversampled balanced dataset

scenarios, we assume a binary outcome $Y \in \{0, 1\}$ with a fixed minority proportion of $\pi_+ = P(Y = 1) = 0.05$ unless otherwise stated, ensuring a high unbalanced class ratio.

Scenario 1: Independent Covariate Case. In the first scenario, let $X \in \mathbb{R}^5$ denote the five-dimensional feature vector. Each X_i is conditionally independent of the other features given the class Y. Specifically, when $Y = 1$, each X_i is drawn from $\mathcal{N}(\mu_{1i}, \sigma_{1i}^2)$, where $\mu_{1i} \sim \text{Unif}(4, 6)$ and $\sigma_{1i} \sim \text{Unif}(1, 2)$. When $Y = 0$, each X_i is drawn from $\mathcal{N}(\mu_{0i}, \sigma_{0i}^2)$, with $\mu_{0i} \sim \text{Unif}(1, 5)$ and $\sigma_{0i} \sim \text{Unif}(2, 5)$.

Scenario 2: Multivariate Correlated Case. In the second scenario, we generate $X \in \mathbb{R}^{10}$ from a mixture of multivariate normal distributions to introduce correlation and multimodality. The minority class ($Y = 1$) consists of two equally probable subpopulations, each characterized by a mean vector μ_{1m} with elements drawn from $\text{Unif}(2, 6)$ to ensure moderate separation. The majority class ($Y = 0$) contains three equally probable subpopulations, each with mean vectors μ_{0m} sampled from $\text{Unif}(0, 4)$. Covariance matrices Σ_{ym} are generated by constructing random matrices A and computing $\Sigma_{ym} = AA^\top$, ensuring positive semi-definiteness and flexible correlations among features.

Scenario 3: High-Dimensional Case. In this scenario, we explore high-dimensional settings by varying the feature dimensionality across experiments, specifically considering $d = 50, 100, 200, 500$. To construct the dataset, we introduce structured sparsity: the first five features are informative, while the remaining $d - 5$ are noise. Informative features are drawn from multivariate normal distributions with class-specific mean vectors sampled from $\text{Unif}(1, 5)$ for the majority class ($Y = 0$) and $\text{Unif}(4, 6)$ for the minority class ($Y = 1$). Covariance matrices are diagonal, with variances from $\text{Unif}(2, 5)$ for $Y = 0$ and $\text{Unif}(1, 2)$ for $Y = 1$. Noise features are independently sampled from a standard normal distribution $\mathcal{N}(0, 1)$, carrying no predictive information.

We adopt a two-level simulation design to incorporate variability in both distribution parameters and sample draws. Specifically, for each scenario, we conduct 5 outer iterations, each time resampling the underlying distribution parameters (e.g., μ, σ, Σ)

and then fixing a test set of size $N_{\text{test}} = 5{,}000$. Within each outer iteration, we perform 500 inner iterations, where in each inner iteration, we generate an independent training set of size $n = 1{,}000$ under the fixed parameters, fit all models, and evaluate on the same test set. This nested procedure captures high-level shifts in the data-generating mechanism while controlling for sampling variability, and we report averaged metrics (with standard deviations) over all runs.

We train five logistic regression models in each inner iteration:

1. **Original Model**: Trained on the unbalanced dataset.
2. **Oversampled Model**: Trained on data where the minority class is randomly oversampled to achieve perfect class balance.
3. **Undersampled Model**: Trained on data where the majority class is randomly undersampled to achieve perfect class balance.
4. **Weighted Model**: Trained on the original unbalanced data using class weights inversely proportional to class proportions.
5. **Synthetic Data Model**: Trained on data where synthetic data generated by cVAE [33] oversamples the minority class to achieve perfect class balance.

Notably, in the high-dimensional case (Scenario 3), we fit debiased logistic regression models with L_1 penalty [18]. Otherwise, standard logistic regression models are fitted. For the synthetic data model, cVAE models are trained on 5,000 auxiliary datasets conditioning on the class label Y – a standard approach in relevant literature [8, 34]. Additional details can be found in Appendix A.

In order to test the efficiency of our proposed correction formulas in Section 4, we adjust the estimated probabilities to account for the altered class distributions due to balancing techniques. The true posterior probabilities $p^{GT} = P(Y = 1 \mid X)$ are computed using the underlying true data-generating distributions and the Bayes' Theorem. We use $\hat{p}^{\text{orig}}$, $\hat{p}^{\text{bal}}$, and $\hat{p}^{\text{adj}}$ to represent the posterior probabilities estimated by models trained on original unbalanced data, on balanced data after applying balancing techniques, and on balanced data with our probability correction solutions, respectively. We evaluate model performance on the fixed test set by calculating the Mean Squared Error (MSE=$\frac{1}{n}\sum_{i=1}^{n}(p_i^{GT} - \hat{p}_i)^2$) and Mean Absolute Error (MAE=$\frac{1}{n}\sum_{i=1}^{n}\left|p_i^{GT} - \hat{p}_i\right|$) between the estimated probabilities and the true probabilities.

For Scenario 1, Table 3 shows that the original model yields the smallest MSE and MAE, indicating that training on the original unbalanced data provides the most accurate probability estimates when no balancing technique is applied. In contrast, all balancing methods introduce noticeable distortion in posterior probability estimation before correction. Among them, undersampling produces the largest errors, likely because removing majority-class observations discards information and increases variability in the fitted model.

After applying the proposed posterior probability correction from Section 4, both MSE and MAE are substantially reduced for all balancing methods. The largest improvements are observed for oversampling and weighting, whose corrected results are very close to those of the original model. The corrected oversampling and weighting results are also nearly identical. This is expected because, under inverse-frequency weighting, the majority-class weight $1/(1-0.05)$ is close to 1, so the weighting scheme

Table 3 Summary of the mean squared error (MSE) and mean absolute error (MAE) before and after applying balancing techniques and probability correction in Scenario 1.

Technique	mean$((p^{GT} - \hat{p})^2)$	mean$(\lvert p^{GT} - \hat{p}\rvert)$	mean$((p^{GT} - \hat{p}^{adj})^2)$	mean$(\lvert p^{GT} - \hat{p}^{adj}\rvert)$
Original	**0.0213**	**0.0617**	—	—
	(0.0054)	(0.0141)		
Oversampled	0.1106	0.2055	0.0255	0.0643
	(0.0345)	(0.0633)	(0.0082)	(0.0175)
Undersampled	0.1299	0.2180	0.0379	0.0788
	(0.0385)	(0.0644)	(0.0179)	(0.0228)
Weighting	0.1107	0.2058	0.0255	**0.0642**
	(0.0345)	(0.0632)	(0.0082)	(0.0175)
Synthetic Data	0.1178	0.2144	**0.0253**	0.0651
	(0.0419)	(0.0762)	(0.0075)	(0.0188)

Table 4 Summary of the mean squared error (MSE) and mean absolute error (MAE) before and after applying balancing techniques and probability correction in Scenario 2

Technique	mean$((p^{GT} - \hat{p})^2)$	mean$(\lvert p^{GT} - \hat{p}\rvert)$	mean$((p^{GT} - \hat{p}^{adj})^2)$	mean$(\lvert p^{GT} - \hat{p}^{adj}\rvert)$
Original	**0.0302**	**0.0589**	—	—
	(0.0057)	(0.0127)		
Oversampled	0.1052	0.1938	0.0349	0.0637
	(0.0366)	(0.0795)	(0.0060)	(0.0137)
Undersampled	0.1343	0.2111	0.0556	0.0857
	(0.0423)	(0.0814)	(0.0285)	(0.0256)
Weighted	0.1054	0.1943	0.0349	0.0638
	(0.0365)	(0.0795)	(0.0060)	(0.0137)
Synthetic Data	0.0753	0.1170	0.0349	**0.0522**
	(0.0369)	(0.0628)	(0.0095)	(0.0180)

largely acts by increasing the relative contribution of the minority class, similar to oversampling.

We further examine different class imbalance ratios to assess whether the proposed correction remains effective across varying imbalance levels. The results in Appendix B show that the correction consistently reduces both MSE and MAE across all balancing methods. However, as the imbalance becomes more severe, undersampling leads to larger errors because more aggressive sample removal causes greater information loss. Overall, these results further support that, although all balancing techniques distort probability estimation to some extent, the proposed correction effectively mitigates this distortion, with oversampling and weighting showing the most stable performance.

For Scenario 2, Table 4 summarizes the results for data generated from a mixture of multivariate normal distributions, which creates a more challenging setting than Sce-

Table 5 Mean Absolute Error (mean on top, standard deviation in parentheses) for each method and each dimension d from Scenario 3.

Metric	Mean ($\lvert p^{GT} - \hat{p} \rvert$)			
Dimension	$d = 50$	$d = 100$	$d = 200$	$d = 500$
Original	0.0660(0.0132)	0.0721(0.0137)	0.0760(0.0165)	0.0678(0.0108)
Over	0.1563(0.0655)	0.1207(0.0400)	0.1039(0.0251)	0.0912(0.0172)
Over+Refit	0.0685(0.0102)	0.0733(0.0081)	0.0748(0.0079)	0.0762(0.0073)
Over+Refit+Adj	0.0495(0.0015)	0.0500(0.0011)	0.0501(0.0011)	0.0503(0.0010)
Under	0.3340(0.0824)	0.3840(0.0902)	0.4242(0.0839)	0.4643(0.0661)
Under+Refit	0.0732(0.0090)	0.0762(0.0078)	0.0782(0.0066)	0.0805(0.0048)
Under+Refit+Adj	0.0501(0.0011)	0.0504(0.0010)	0.0506(0.0009)	0.0507(0.0007)
Weighted	0.1612(0.0658)	0.1258(0.0425)	0.1068(0.0262)	0.0943(0.0183)
Weighted+Refit	0.0679(0.0107)	0.0722(0.0089)	0.0741(0.0084)	0.0754(0.0079)
Weighted+Refit+Adj	0.0494(0.0015)	0.0499(0.0012)	0.0501(0.0011)	0.0502(0.0011)
Synthetic Data	0.2984(0.1160)	0.3294(0.1219)	0.3391(0.1556)	0.5071(0.0102)
Synthetic Data+Refit	0.0671(0.0108)	0.0709(0.0099)	0.0749(0.0096)	0.0841(0.0030)
Synthetic Data+Refit+Adj	0.0493(0.0016)	0.0497(0.0015)	0.0501(0.0013)	0.0510(0.0006)

nario 1 due to correlation and multimodality. As expected, both MSE and MAE are higher across all methods. Undersampling again produces the largest errors, suggesting that information loss is especially harmful in multimodal settings. Nevertheless, the proposed correction remains effective and substantially reduces both MSE and MAE for all balancing methods. Among the corrected methods, synthetic-data oversampling achieves the smallest MAE, while oversampling and weighting also perform competitively. These findings show that the proposed correction remains useful even when the data-generating mechanism is more complex.

For Scenario 3, Table 5 reports the mean absolute error across different feature dimensions for the high-dimensional setting. Two patterns are clear. First, refitting substantially reduces the error for all balancing methods, indicating that it effectively alleviates the shrinkage-induced distortion caused by penalized logistic regression. Second, after refitting, the proposed probability adjustment further improves calibration and consistently yields the smallest MAE within each balancing strategy.

After the full correction pipeline, the adjusted methods perform very similarly across dimensions, with MAE values close to 0.05 in most cases. Among them, the synthetic-data and weighted versions are slightly better in several settings, while undersampling remains less stable before correction because discarding observations is particularly harmful in high-dimensional problems. Overall, these results show that combining refitting with the proposed posterior correction effectively addresses both regularization-induced shrinkage and balancing-induced probability distortion.

Table 6 Summary of datasets, sample sizes, and class proportions.

Datasets	Sample Size	π_+	π_-
AIDS 175	2139	24.35%	75.65%
letter-vowel	20000	19.39%	80.61%
page blocks	5473	10.23%	89.77%
Cirrhosis Patient Survival	418	5.98%	94.02%
letter-a	20000	3.94%	96.06%
car evaluation-vgood	1728	3.76%	96.24%
cover type-4	581012	0.47%	99.53%
credit card	284807	0.17%	99.83%

6 Applications

To evaluate the effectiveness of our proposed posterior probability correction and threshold adjustment methods in real-world scenarios, we conduct a series of experiments using datasets from the UCI Machine Learning Repository [42] and other publicly available sources. We select eight datasets, each with different degrees of class imbalance, to ensure a robust evaluation of our methods. Table 6 summarizes the datasets, their sample sizes, and the class distributions used in our study. Each dataset is split into training and testing sets with a 70/30 ratio. In each repetition, 80% of the training set is randomly selected for model fitting. Repeating this procedure over 100 iterations provides a stable assessment of model performance under varying training samples.

The experiments compare logistic regression models trained with different balancing strategies, both with and without the proposed probability correction and threshold adjustment. Specifically, we consider the following models:

1. **Baseline model:** logistic regression trained on the original unbalanced dataset, using the standard decision threshold of 0.5.
2. **Balanced models:** logistic regression trained using random oversampling, random undersampling, class weighting, and synthetic data augmentation, respectively, each with decision threshold 0.5.
3. **Corrected models:** the balanced models in Step 2 after applying the proposed posterior probability correction and replacing the default threshold 0.5 with the class-prior threshold π_+, as described in Section 4.

Details of these models are given in Appendix A. Unlike in the simulation studies, the synthetic data generator (cVAE) is trained directly on the real training data rather than on auxiliary datasets. Details of above models can be found in Appendix A. Notably, unlike simulation studies, the synthetic data generators (i.e., cVAE) are trained on training data instead of auxiliary datasets.

To evaluate model performance comprehensively, we select metrics based on two core objectives: (1) classification effectiveness and (2) probability calibration quality. We use AUC [43–45] and G-Mean [20] to evaluate the classification performance. For evaluating the quality of probability estimation, we adopt Expected Calibration

Table 7 Comparison of metrics across original, oversample, undersample, and weighting models for page blocks, credit card, and car evaluation-vgood datasets.

Dataset	Model	Adjustment	AUC	G-Mean	ECE	Log Loss	Brier Score(%)
Page Blocks	Original	—	0.9511	0.7514	0.1711	0.1673	4.2884
	Oversample	Unadjusted	0.9718	0.9019	0.2272	0.2868	5.9543
		Adjusted	0.9718	0.9019	0.1417	0.1889	3.9880
	Undersample	Unadjusted	0.9717	0.8994	0.2112	0.3383	6.2656
		Adjusted	0.9717	0.8994	0.1622	0.2356	4.3674
	Weighting	Unadjusted	0.9724	0.9017	0.2216	0.2881	5.9352
		Adjusted	0.9724	0.9017	0.1443	0.1897	4.0196
	Synthetic	Unadjusted	0.9485	0.8702	0.2530	0.2190	5.0772
		Adjusted	0.9485	0.8702	0.0886	0.1708	3.8451
Credit Card	Original	—	0.9698	0.7985	0.2035	0.0044	0.0686
	Oversample	Unadjusted	0.9721	0.9408	0.4682	0.1075	2.3609
		Adjusted	0.9721	0.9408	0.3340	0.0100	0.0859
	Undersample	Unadjusted	0.9732	0.9389	0.4808	0.1527	3.4197
		Adjusted	0.9732	0.9805	0.0164	0.1895	0.1174
	Weighting	Unadjusted	0.9721	0.9408	0.4682	0.1076	2.3630
		Adjusted	0.9721	0.9408	0.3331	0.0100	0.0850
	Synthetic	Unadjusted	0.9683	0.9353	0.4645	0.0740	1.2355
		Adjusted	0.9683	0.9353	0.2105	0.0127	0.0756
Car Evaluation-vgood	Original	—	0.9509	0.2216	0.2748	0.1037	3.2682
	Oversample	Unadjusted	0.9489	0.9029	0.3791	0.3012	8.8231
		Adjusted	0.9489	0.9029	0.3134	0.1146	3.7206
	Undersample	Unadjusted	0.9357	0.8854	0.3974	0.3802	12.2215
		Adjusted	0.9357	0.8854	0.2487	0.1132	3.4635
	Weighting	Unadjusted	0.9488	0.9060	0.3817	0.3016	8.9950
		Adjusted	0.9488	0.9060	0.3145	0.1117	3.6329
	Synthetic	Unadjusted	0.9336	0.8724	0.4208	0.4201	13.7229
		Adjusted	0.9336	0.8724	0.1876	0.1079	3.3008

Error (ECE), Log Loss, and Brier Score. These metrics are standard in assessing how well estimated probabilities align with actual outcomes [19, 46]. ECE approximates the gap between confidence and accuracy across prediction bins, Log Loss penalizes probabilistic errors exponentially, and the Brier Score captures squared deviations between predicted and actual class labels.

Table 7 summarizes the performance of the original, oversampled, undersampled, weighted, and synthetic data models, both before and after probability correction and threshold adjustment, across three datasets described in Table 6. Results on remaining datasets reveal similar conclusions and can be found in Appendix C. The key findings can be summarized as follows:

Improvement in Classification Accuracy Metrics: Balancing techniques generally improve classification accuracy compared to the baseline logistic regression model. For example, in the page block dataset, the baseline model achieves an AUC of 0.9511 and a G-Mean of 0.7514, while random oversampling improves these to 0.9718 and 0.9019, respectively. Similar improvements are observed across other balancing techniques and other datasets, where G-Mean improves by at least 0.15. Occasionally, balancing techniques may lead to a slightly decreased AUC. For example, in the car evaluation-vgood dataset, the baseline AUC is 0.9509 and drops to 0.9489 after oversampling. This decline in performance could be attributed to several factors, such as overfitting, reduced generalization, and noise introduced by random sampling.

Our posterior probability correction and threshold adjustments consistently maintain the high accuracy levels for models trained on rebalanced datasets. This consistency indicates that our adjustments preserve the classification capability brought by balancing techniques, so it does not adversely affect the models' ability to distinguish between classes.

Probability calibration. Models trained on the original unbalanced data generally exhibit better calibration than unadjusted balanced models, as reflected by lower ECE, log loss, and Brier score. For example, on the car-evaluation dataset, the baseline model achieves an ECE of 0.2748, a log loss of 0.1037, and a Brier score of 3.2682%, whereas the unadjusted oversampled model has substantially worse calibration, with values of 0.3791, 0.3012, and 8.8231%, respectively.

The proposed correction substantially improves calibration for balanced models across datasets and balancing methods. In the same example, the corrected oversampled model reduces ECE, log loss, and Brier score to 0.3134, 0.1146, and 3.7206%, bringing its calibration much closer to that of the baseline model. These results show that the proposed adjustment effectively recovers posterior probabilities that better reflect the original data distribution.

Although the corrected models do not always fully match the calibration performance of models trained on the original unbalanced data, the remaining gap is typically small. More importantly, the correction allows probability estimation and classification to be carried out within the same balanced model, avoiding the inconsistency that may arise when separate models are used for these two tasks. Therefore, when both strong classification performance and reliable probability estimates are needed, the corrected balanced models provide a practical and principled solution.

Robustness across balancing methods and datasets. The proposed correction is computationally simple, as it is a post-hoc monotone transformation of the predicted probabilities and does not require retraining the underlying model. This makes it easy to deploy across different balancing strategies and learning pipelines.

Across all real datasets, the correction consistently improves probability calibration for models trained on balanced data. The improvements are reflected in lower ECE, log loss, and Brier score, showing that the adjustment effectively mitigates the systematic distortion introduced by class rebalancing. Methods that retain more information from the original data, such as weighting and oversampling, generally benefit the most and often approach the calibration performance of the baseline model trained on the original data. Undersampling also benefits from correction, but its residual errors are

typically larger because discarding observations increases variance. Overall, these findings suggest that the proposed correction addresses a common source of bias shared across balancing methods, namely the distortion of effective class priors.

7 Limitation and Future Work

Dependency on the Classification Performance. Our method relies on an initial classification model to estimate the posterior probability. If the original model poorly distinguishes between classes, our calibration approach cannot correct fundamental classification inaccuracies. Future works could explore hybrid approaches that integrate calibration with incremental or adaptive model training to improve classification performance even when the initial model is suboptimal.

Limitations of Threshold Adjustment. Our threshold adjustment approach assumes accurate knowledge of the true class prior (π_+). In practice, this prior may be unknown or vary dynamically, potentially leading to suboptimal classification performance. Future research could focus on adaptive methods for estimating class priors to enhance robustness across diverse and evolving tasks.

Validation of Assumptions. Our study assumes the standard binary classification scenario with stable underlying class distributions and clearly defined priors. These assumptions were validated using commonly referenced, well-studied benchmark datasets known for significant class imbalance from the UCI Machine Learning Repository [42]. Despite this, we acknowledge potential challenges from dissenting perspectives, particularly in dynamic or streaming data scenarios where the distribution of classes changes over time, or in multi-class settings. These real-world complexities may challenge the direct applicability of our calibration approach, suggesting a valuable direction for future research.

8 Conclusion

In this paper, we explore the challenges of training classifiers on unbalanced datasets and the potential distortions introduced by common balancing techniques, such as oversampling, undersampling, and class weighting. Although these methods improve classification performance for the minority class, they often distort the posterior probability distribution, leading to biased probability estimates that do not accurately reflect real-world conditions. Our research demonstrates that models trained on artificially balanced datasets may offer improved classification metrics but introduce biases that impair probability estimation. We show that the probability distortions from balancing techniques lead to misaligned probability estimations, as illustrated in our toy example and simulated studies.

We note that using separate models, one for probability estimation (trained on the original dataset) and one for classification (trained on the balanced dataset), is not feasible due to potential inconsistencies in these outputs. To address these issues, we propose correction methods to adjust the posterior probability and decision thresholds

in models trained with balancing techniques. Our approach aims to maintain the classification benefits of balancing methods while restoring the integrity of probability estimates, allowing models to generalize more effectively to real-world data. Both simulated and real-data results confirm that our correction techniques significantly alleviate the biases caused by balancing techniques, without sacrificing classification accuracy.

In summary, this paper highlights the need for careful consideration of probability calibration when applying balancing techniques to unbalanced datasets. Our proposed correction methods contribute to a more reliable framework for handling unbalanced classification, offering both improved classification metrics and more accurate probability estimates. Future work may focus on extending these corrections to other classification algorithms and exploring their application in more complex real-world scenarios to further validate their generalizability and effectiveness.

Declarations

Conflicts of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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Appendix A Details of Simulation and Applications

In low-dimensional settings, we fit standard logistic regression models. To handle high-dimensional data, we employ L_1-penalized logistic regression. Since regularization induces bias in probability estimation, we adopt a refitting strategy to correct the estimated posterior probabilities. This approach was first introduced by Platt [47] and later refined by Zhang et al. [18]. Specifically, after fitting a high-dimensional regularized classifier, we obtain its classification score $f(x)$ and refit a one-dimensional logistic regression model using $f(x)$ as the only covariate. The calibrated posterior probability is then estimated as

$$P(Y = 1 \mid f(x)) = \frac{1}{1 + \exp(Af(x) + B)},$$

where the parameters A and B are estimated using a held-out validation set.

Standard balancing techniques are applied to the simulated datasets. For the weighted model, class weights are automatically assigned based on the inverse class frequencies implied by the response variable Y. For resampling-based methods, includ-

Table 8 Comparison of MSE and MAE before and after correction under varying class imbalance ratios (π_+) in Scenario 1. The values in each cell represent the mean, with the standard deviation provided below it.

π_+	Original	Oversampled		Undersampled		Weighted	
	MSE	MSE (Unadj.)	MSE (Adj.)	MSE (Unadj.)	MSE (Adj.)	MSE (Unadj.)	MSE (Adj.)
0.10	0.0108	0.0260	0.0105	0.0297	0.0114	0.0260	0.0106
	(0.0022)	(0.0067)	(0.0028)	(0.0080)	(0.0042)	(0.0063)	(0.0026)
0.05	0.0104	0.0355	0.0096	0.0450	0.0105	0.0356	0.0097
	(0.0018)	(0.0108)	(0.0023)	(0.0159)	(0.0044)	(0.0105)	(0.0025)
0.01	0.0080	0.0451	0.0065	0.0954	0.0087	0.0453	0.0065
	(0.0013)	(0.0245)	(0.0020)	(0.0475)	(0.0037)	(0.0234)	(0.0020)

ing random oversampling and undersampling, the training data are adjusted so that the positive and negative classes are equally represented.

In addition to these classical balancing strategies, we also consider a synthetic-data-based balancing approach using a cVAE. Specifically, we train a cVAE on an auxiliary learning dataset of size 5,000 drawn from the same data-generating process, conditioning on the class label Y. The trained cVAE is then used to generate additional minority-class observations, which are combined with the original training data to form a balanced dataset. A logistic regression model is subsequently trained on this cVAE-augmented dataset. As discussed in Section 4, we apply the same posterior probability correction derived for random oversampling to the cVAE-based method.

The cVAE consists of an encoder and a decoder network. The encoder maps the input data to an 8-dimensional latent space through three fully connected layers with widths (64, 64, 8), each followed by a ReLU activation function. The decoder mirrors this architecture, projecting the latent representation back to the original data dimension using three fully connected layers with widths [8, 64, 64] and ReLU activations. The cVAE is trained by optimizing the objective described in Sohn et al. [33], with the Kullback–Leibler divergence term weighted by 0.5. Optimization is performed using the Adam optimizer with a learning rate of 10^{-3} and gradient norm clipping at 5. The batch size is set to 512, and the model is trained for up to 80 epochs. To prevent overfitting, early stopping is employed if the training loss fails to decrease by at least 10^{-4} over 8 consecutive epochs.

The real data applications follow the same modeling framework as the low-dimensional simulation settings (Scenarios 1 and 2). Unlike in the simulation studies, however, the cVAE models for real data are trained directly on the available training data rather than on auxiliary datasets.

Appendix B Extended Simulation

As an extension, we vary the class imbalance ratio to examine the behavior of the proposed correction across different degrees of imbalance. The detailed results are reported in Table 8. Across all balancing strategies, the correction consistently reduces

Table 9 Performance comparison across five additional real-world datasets.

Dataset	Model	Adjustment	AUC	G-Mean	ECE	Log Loss	Brier Score (%)
Letter - "A"	Original	—	0.9904	0.9106	0.0954	0.0377	0.9834
	Oversample	Unadjusted	0.9895	0.9622	0.3710	0.1431	3.7585
		Adjusted	0.9895	0.9622	0.0842	0.0501	1.3813
	Undersample	Unadjusted	0.9886	0.9574	0.3813	0.1591	4.2273
		Adjusted	0.9886	0.9574	0.0840	0.0539	1.4570
	Weighting	Unadjusted	0.9895	0.9625	0.3707	0.1430	3.7537
		Adjusted	0.9895	0.9625	0.0846	0.0500	1.3807
	Synthetic	Unadjusted	0.9869	0.9357	0.2760	0.0948	1.9953
		Adjusted	0.9869	0.9357	0.1272	0.0650	1.4019
Letter - "AEIOU"	Original	—	0.7495	0.3419	0.2034	0.4369	14.0127
	Oversample	Unadjusted	0.7581	0.6985	0.2895	0.6104	20.5439
		Adjusted	0.7581	0.6985	0.2472	0.4417	14.2953
	Undersample	Unadjusted	0.7574	0.6981	0.2885	0.6126	20.6116
		Adjusted	0.7574	0.6981	0.2485	0.4427	14.3376
	Weighting	Unadjusted	0.7583	0.6989	0.2897	0.6105	20.5382
		Adjusted	0.7583	0.6989	0.2474	0.4417	14.2948
	Synthetic	Unadjusted	0.7537	0.6882	0.2641	0.6238	20.5796
		Adjusted	0.7537	0.6882	0.1744	0.4563	14.5221
Cover type	Original	—	0.9944	0.5747	0.0719	0.0106	0.3040
	Oversample	Unadjusted	0.9942	0.9811	0.4649	0.0883	2.3839
		Adjusted	0.9942	0.9811	0.1968	0.0130	0.3975
	Undersample	Unadjusted	0.9937	0.9799	0.4666	0.0958	2.5288
		Adjusted	0.9937	0.9799	0.2121	0.0138	0.4214
	Weighting	Unadjusted	0.9944	0.9809	0.4646	0.0870	2.3528
		Adjusted	0.9944	0.9809	0.1938	0.0128	0.3922
	Synthetic	Unadjusted	0.9907	0.9022	0.3981	0.0582	1.1919
		Adjusted	0.9907	0.9022	0.2503	0.0234	0.5084
AIDS-175	Original	—	0.8692	0.7343	0.0810	0.3506	10.3226
	Oversample	Unadjusted	0.8781	0.8171	0.1279	0.3908	11.6993
		Adjusted	0.8781	0.8171	0.0916	0.3514	10.5360

both squared and absolute errors in posterior probability estimation. As the imbalance becomes more severe, the errors associated with undersampling increase, reflecting the impact of information loss due to aggressive sample removal. In contrast, oversampling and weighting exhibit more stable error patterns across different imbalance levels after correction.

Table 9 continued

Dataset	Model	Adjustment	AUC	G-Mean	ECE	Log Loss	Brier Score (%)
	Undersample	Unadjusted	0.8761	0.8139	0.1380	0.4010	12.0829
		Adjusted	0.8761	0.8139	0.0866	0.3547	10.6496
	Weighting	Unadjusted	0.8790	0.8181	0.1273	0.3900	11.6728
		Adjusted	0.8790	0.8181	0.0928	0.3496	10.4948
	Synthetic	Unadjusted	0.8690	0.8134	0.1922	0.4395	12.9598
		Adjusted	0.8690	0.8134	0.0723	0.3786	10.7719
Cirrhosis Patient Survival	Original	—	0.7120	0.3178	0.2729	0.2485	6.6918
	Oversample	Unadjusted	0.6935	0.6027	0.4128	0.6211	19.2785
		Adjusted	0.6935	0.6027	0.3461	0.3026	7.5259
	Undersample	Unadjusted	0.7056	0.6481	0.4314	0.8595	27.9281
		Adjusted	0.7056	0.6481	0.3306	0.2924	8.5573
	Weighting	Unadjusted	0.7014	0.6257	0.4106	0.6065	19.2912
		Adjusted	0.7014	0.6257	0.3208	0.2827	7.1588
	Synthetic	Unadjusted	0.5533	0.5580	0.4405	0.8075	27.7758
		Adjusted	0.5533	0.5580	0.3141	0.2894	7.5474

Appendix C Additional Applications

We evaluate the proposed methods using additional datasets listed in Table 6. Consistent with the main results, the findings in Table 9 demonstrate that (i) balancing techniques generally improve predictive accuracy, and (ii) our correction methods substantially enhance posterior probability estimation across a range of balancing approaches.

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